GamersDen: A Smart Platform for Gamers and Gaming Centres

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Abstract

Gaming and esports centres provide social entertainment spaces where customers can reserve consoles and play with friends. However, information about venue locations, facilities, prices and available time slots is often scattered across social-media pages, messaging applications and phone calls. This can make it difficult for customers to compare centres and confirm a suitable session, while venue operators may face challenges maintaining accurate reservations and preventing duplicate bookings. GamersDen is proposed as a web-based platform that connects customers with nearby PlayStation gaming centres and provides a centralized workflow for venue discovery and gaming-session booking. Customers can search for centres by location, review venue details, facilities, pricing and contact information, and select an available date and time slot. Venue operators can maintain centre information, update availability and review reservations through an organized interface. The proposed system uses HTML, CSS, Java and React for the presentation layer, Django for server-side application logic, authentication, venue search and booking validation, and MySQL for persistent storage of user, gaming-centre, facility, slot and booking records. During a booking request, the backend checks slot availability before saving the reservation, helping reduce conflicting or duplicate bookings. The design also emphasizes input validation, secure password hashing, access control for venue-management functions and safe database operations. The evaluation plan covers venue-discovery relevance, booking reliability, availability consistency, response time and usability. Testing is proposed for common workflows and repeated or concurrent booking attempts; numerical evaluation results are not claimed in this paper. GamersDen aims to reduce dependence on manual communication, improve booking transparency and provide customers and gaming-centre operators with a more convenient way to

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manage gaming sessions. Future enhancements may include map-based distance ranking, online payments, notifications, ratings and reviews, promotions, analytics and mobile applications.

Keywords— PlayStation booking, gaming centres, location-based services, React, Django, MySQL, online reservation

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